

SRINIVAS DHARPALLY

AI/ML Engineer | Geospatial Computer Vision & LiDAR | Full-Stack

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Summary

AI/ML engineer building production computer-vision pipelines on drone, satellite and LiDAR data, and the GIS platforms that serve them. Team lead for LiDAR deep-learning R&D (PointNet, PointNet++, KPConv); shipped 4 drone-CV and 2 statewide satellite models; own ~80% of a capital-region GeoPortal backend.

Technical Skills

ML / Deep Learning: PyTorch, UNet++ multi-head segmentation, PointNet/PointNet++/KPConv (LiDAR), SAM3, LangGraph, RAG, OpenAI SDK
Geospatial: PostGIS, GeoServer, GeoPandas, Rasterio/GDAL, Shapely, pgRouting, CloudCompare, QGIS, Planet imagery
Languages & Backend: Python, C++, Java, JavaScript/TypeScript, SQL; Django, REST APIs, PostgreSQL, Elasticsearch, Keycloak (RBAC), Streamlit
Frontend, Mobile & DevOps: React, Next.js, React Native (Expo); Docker, Jenkins CI/CD, Nginx, Monit, Linux, Git

Experience

R&D Engineer I – AI/ML & Geospatial

Jan 2025 – Present

VassarDigital.ai | Team Lead, LiDAR R&D | Python, PyTorch, Django, PostGIS, Docker

- **LiDAR railway asset extraction (Team Lead):** implemented **PointNet, PointNet++, KPConv** end-to-end (data, training, inference); extracted centrelines, rails, dead-ends, poles; classified Main/Loop/Siding/Turnout tracks.
- **Railway yard remodelling (ESP generation, 100%):** geometry engine for turnouts (switch/lead/crossing/connection; straight, single- and reverse-curve) and **9 loop-line layouts** per **IRSOD/IRPWM**.
- **Drone & satellite CV:** 4 production pipelines (dual-head UNet++ parcels, buildings + DEM heights, triple-head roads, SAM3 green cover) and 2 statewide Planet models (aqua **88%**, built-up **90%**), dataset to portal API.
- **GeoPortal backend (~80%, 21 modules)** for the Amaravati capital region; migrated Cassandra → PostgreSQL/PostGIS, improving spatial queries **60%** and creating a single data source.
- **View-Creation Module:** config-driven shapefile → N-level aggregations → GeoServer layers; new dashboards with zero backend effort, onboarding **5–6 h → 10 min**.
- **Spatial analysis toolkit:** query over **520+ layers**, buffer/overlay, nearest-neighbour via **pgRouting Dijkstra**, drawing tool, 7-format GIS export; graph-based **Road-Hit-Plots** workflow with width-aware buffers.
- **Search, citizen portal & access:** Elasticsearch from scratch; OTP-authenticated LPS farmer portal with plot search and PII-aware masking; Keycloak RBAC; metadata management with shapefile version revert.
- **DevOps & security:** Jenkins CI/CD (**–80%** deployment effort), **Night Watcher** (Monit) self-healing alerts, security-audit remediation (Nginx, backend, users), GeoServer PII encryption; PRRD Elasticsearch setup.

Full-Stack Developer (Freelance)

Sep 2025 – Jan 2026

Amplify Insights | Python, Django, Next.js, PostgreSQL, Linux

- **Media Scan:** news aggregation, Telugu/English summaries, 50+ channel YouTube analytics; reports **4 h → 1 min**.
- **Voter Analytics:** PDF extraction with AI-assisted regex, upload → extract → validate → export workflow with RBAC; district/constituency insights.
- **Action Management & KCR Cup:** action tracking with audit trails and notifications; cricket platform with live scoring.

Full-Stack & AI Engineer (Freelance)

Jun 2025 – Dec 2025

ShapeCoach.ai | Django, React Native (Expo), LangGraph, RAG, OpenAI, PostgreSQL, Razorpay

- Owned a cross-platform AI fitness app (Android, iOS, Web); Razorpay orders, signature verification and subscriptions.
- **LangGraph** multi-agent system with RAG over 900+ exercises generating monthly workout and 7-day diet plans; OpenAI Agent SDK chatbot.

Education

Rajiv Gandhi University of Knowledge Technologies, Basar

2021 – 2025

B.Tech, Computer Science & Engineering

CGPA 9.18

RGUKT Basar

2019 – 2021

Pre-University Course

CGPA 8.97

Z.P.H.S Pachala Nadkuda

2018 – 2019

Secondary School Certificate (SSC)

CGPA 10.0